
FORKSERVE: PREFILL PRUNING AND DISAGGREGATED KV FOR AGENTIC LLM SERVING

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ABSTRACT

Agentic serving expands one trunk of length L into k residuals and retains one spine. A prefix cache copies L once per child on the first fan-out, because a block hash exists only after the tokens do. Disaggregation then transfers every page that was computed, including residuals the harness aborts. Decode-time pruning (ESC, Speculative Rejection, DPTS) scores a child only after that prefill.

ForkServe charges residual i only for the suffix absent from the radix index, and only when rejection is not yet admissible. Let m_i be the longest full-page prefix of ρ_i already published. If $m_i = \ell_i$, no kernel runs and nothing is transferred. If $0 < m_i < \ell_i$, prefill and transfer cover $\ell_i - m_i$. If $m_i = 0$, a repeated loop is withheld before the kernel; every other residual is prefilled, and a score below τ continues past a t_0 -token probe only when the generated prefix meets τ . The connector inserts exactly that work on the retained set. The trunk is an alias, so L is absent from both the kernel and the transfer, and abort leaves the spine.

On Qwen3-8B GSM8K ($n=16$, $D=512$), $\tau_{\text{pre}}=0.15$ ahead of ESC, Speculative Rejection, or DPTS admits 12 of 16 residuals, cuts computed prefill from 5008 to 3848 tokens (-23%), holds 16/16 any-finished at $k=16$, and reduces ESC time from 17.05 s to 14.14 s. $\tau_{\text{pre}}=0.45$ reaches 10.93 s at the same 16/16 and loses two problems at $k=4$ and $k=8$. On Qwen3-8B MATH-500, withholding every residual with score below τ drops any-finished accuracy from 20/32 to 13/32 at $k=16$; continuing after a 128-token probe restores 20/32 and cuts end-to-end time from 104.1 s to 72.2 s. On Qwen3-14B GSM8K the spine matches prefix-cache accuracy at $0.72\times$ the median peak KV.

1 INTRODUCTION

LLM serving began as a single-turn problem: a prompt arrives, the engine prefills, and tokens are streamed [47, 17, 53]. Production agents—coding assistants [44, 39], computer-use systems [55], and multi-agent workflows [42, 14]—execute programs of LLM calls interleaved with tools [46, 50]. Tree-of-Thoughts and MCTS expand sibling thoughts that share a prefix and diverge in a short residual [45, 26]. The serving unit is a *branching execution*: one trunk of length L , k residuals ℓ_i , and one committed spine.

Three families of systems apply to that execution, and each incurs branch cost at a fixed stage (§2). Automatic prefix caching (APC) [17] and RadixAttention [53] reuse a trunk after a full block has been published, so a same-batch fan-out still materializes k copies of L . Prefill–decode disaggregation [54, 32] isolates TTFT from inter-token latency and then transfers every page the prefiller computed, aborted residuals included. Decode-time pruning scores

a child from tokens that are already resident: ESC [21] waits for an agreeing window, Speculative Rejection [31] waits for a partial-reward horizon, and DPTS [19] waits for a mini-step. In each case the residual has already been prefilled. Program-, session-, and workflow-level schedulers [25, 16, 20, 43, 36, 23, 48] raise the scheduling unit above the request and still issue the next call as a unique continuation. Once KV retention has removed trunk recompute [20, 43], the remaining TTFT term is residual-suffix prefill of children the harness can name before the observation returns.

This work. ForkServe is one admission rule on a shared trunk (Figure 1, §3.4). For residual ρ_i , m_i is the longest page-aligned prefix already in the radix index. That length fixes both the prefill W_i and the transfer X_i : a full hit moves nothing, a partial hit moves the miss, a repeated loop ($i \in \Lambda$) is withheld, and every other miss is prefilled. A score below τ extends past t_0 tokens only if the probe meets τ . Index 0 stays in $\mathcal{K}$, so the connector carries the miss tail of the retained set rather than the k -way trunk [54, 32]. The trunk is aliased at `fork` (§3.2), which is why L is absent from W_i and from X_i ; abort leaves $M_{\text{spine}} = b(L + \ell_*)$. Decode

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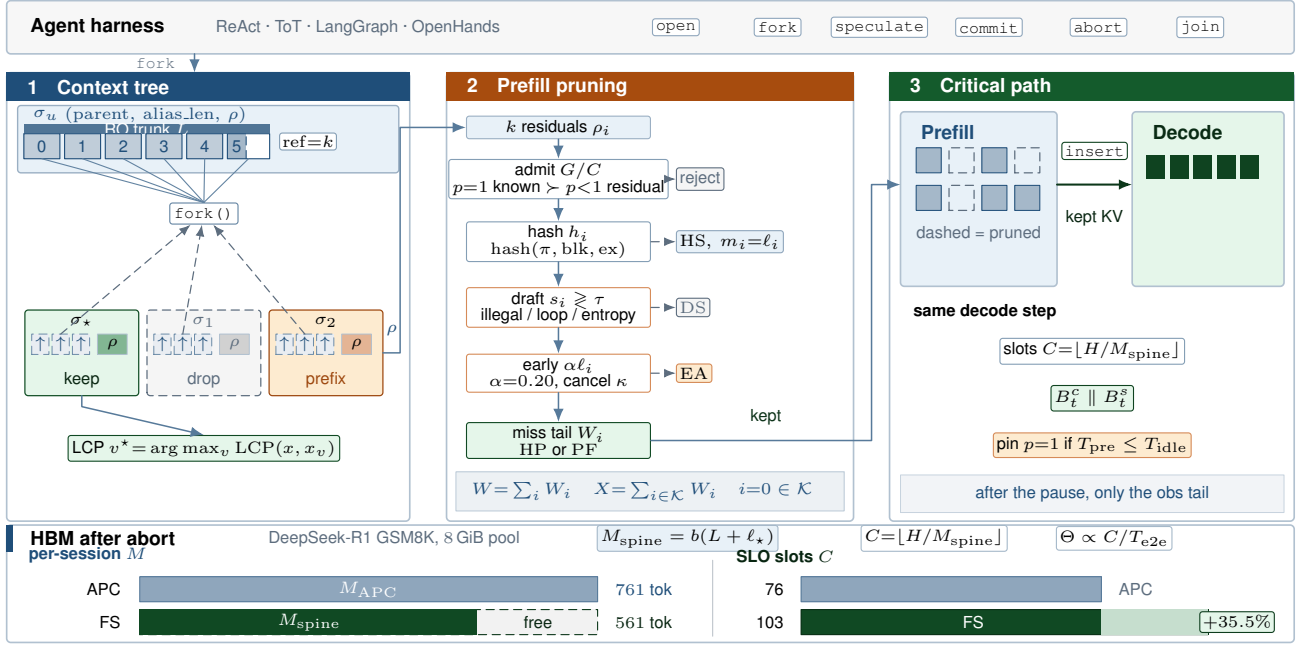


Figure 1. ForkServe architecture. (1) σ_u aliases one trunk; `fork` is $O(1)$; (3) binds the winner. (2) One action a_i ((5)), fixed by the page-aligned match m_i . $m_i = \ell_i$ issues no kernel; $0 < m_i < \ell_i$ prefills the miss; $i \in \Lambda$ is withheld; every other miss is prefilled. X sums W_i over $\mathcal{K}$. (3) $C = \lfloor H/M_{\text{spine}} \rfloor$. After abort the live set is the spine, and the same pool admits 76 $\rightarrow$ 103 sessions.

slots are $C = \lfloor H/M_{\text{spine}} \rfloor$, and any further prefill uses the budget left after committed decode (§3.5). (3) binds the observed prompt; speculative tokens stay out of the sampler.

§4 measures that rule. On Qwen3-8B GSM8K, $\tau_{\text{pre}}=0.15$ in front of ESC, Speculative Rejection, or DPTS admits 12 of 16 residuals, cuts computed prefill by 23% (5008 to 3848), keeps 16/16 any-finished at $k=8$ and $k=16$, and reduces ESC time from 17.05 s to 14.14 s; $\tau_{\text{pre}}=0.45$ is faster at $k=16$ (10.93 s, still 16/16) and loses two problems at smaller fan-out (Tables 8 and 9). On Qwen3-8B MATH-500, withholding every score below τ drops any-finished accuracy from 20/32 to 13/32 at $k=16$; a 128-token probe restores 20/32 and cuts end-to-end time from 104.1 s to 72.2 s (Table 7). Across three turns, 48 of 192 residuals are prefilled and expand time is 7.9–10.6 $\times$ below APC (Tables 11 and 12). The Qwen3-14B spine matches APC accuracy at 0.72 $\times$ median peak KV, and six model families stay in 0.72–0.75 $\times$ ($n=40$).

2 BACKGROUND AND MODEL

2.1 Prefill, Decode, and Idle

Autoregressive inference has two mechanically different phases [47, 54, 32]. *Prefill* consumes the prompt in parallel and is compute-bound; *decode* emits one token (or a small chunk) per step and is memory-bandwidth bound.

PagedAttention [17] stores KV in non-contiguous pages; RadixAttention [53] indexes pages by token prefix so concurrent requests that share a system prompt hit a common trunk. Two consequences follow for agent workloads. First, next-turn TTFT is the prefill of tokens that miss the cache, plus queuing. Second, a memory-bound decode batch rarely saturates the tensor cores that a concurrent chunked prefill can use [2]. For agents, the prompt of turn $i+1$ is a delta on turn i , applied after a tool interval of highly variable length [20, 43].

2.2 Traditional Methods on a Fan-out

We group under *traditional methods* the mechanisms a production stack already applies to one fan-out: prefix caching, hash-indexed prefill, prefill–decode disaggregation, and decode-time pruning. Each either shares the trunk or rejects a child, and the stage of that operation is fixed. Table 1 records the stage.

First-expand cost. APC indexes a full block by $h = \text{hash}(\text{parent}, \text{tokens}, \text{extra})$ after those tokens exist [17]; RadixAttention indexes the same prefix [53]. A same-batch expansion therefore misses and clones k trunks, $W_{\text{APC}} = kL + \sum_i \ell_i$, while a later replay of x_* is a hit ($W=0$). Baseline disaggregation [54, 32] transfers that span,

$$X_{\text{PD}} = kL + \sum_i \ell_i, \quad (1)$$

Table 1. Where each traditional method pays for one fan-out of k residuals on a trunk of length L . “First expand” is the GPU prefill issued before any child has published a block. “Rejection” is the earliest point at which a non-winner can leave. The ForkServe row is the action of §3.4.

Method	First expand	Rejection	Connector
Recompute	k clones of L	end of the request	none
APC, radix, hash prefill	k clones of L	LRU after a block is published	none; a later replay of the winner is a hit
P/D disaggregation	k clones of L	after insert	$kL + \sum_i \ell_i$
ESC, Speculative Rejection, DPTS	every residual	after a generated prefix	KV of every prefilled child
ForkServe	one alias of L	loop before the kernel; other low scores after a t_0 probe	miss tail of $\mathcal{K}$

Table 2. Notation for Definition 1 and for (5)–(7).

Symbol	Domain	Meaning
$L, \ell_i, \ell_\star$	$\mathbb{N}$	trunk; residual i ; committed residual
ρ_i, x_u	tokens	residual of child i ; committed trunk
h_i	digest	$\text{hash}(\text{parent}, \rho_i[0:P], \text{extra})$
m_i	tokens	full-page prefix of ρ_i in H , steps of P
s_i, s'_i, τ	$[0, 1]$	score of ρ_i ; score of the probe y_i ; threshold
t_0, D, d_i	tokens	probe length; decode budget; length assigned to i
α	$(0, 1]$	early-abort prefix fraction (0.20)
p_i, q_i	$[0, 1]$	admit probability; residual-prefix hit rate
$a_i \in \mathcal{A}$	$\mathcal{A}$	HS, HP, DS, EA, PF ((5))
Λ, F_i	set, $\{0, 1\}$	repeated-loop index set; early-abort predicate
$\mathcal{K}$	$\subseteq [k]$	residuals that remain after admission
W_i, X_i	tokens	GPU miss tail; connector payload
b, P	bytes, tok	KV bytes per token; page size
$B_t = B_t^c + B_t^s$	tokens	committed and speculative budget on tick t

so $X_{PD} = W_{APC}$ on the first expand. ESC [21], Speculative Rejection [31], and DPTS [19] evaluate a child only after its prefill and a generated prefix, hence after W and, under disaggregation, after X .

2.3 Fan-out Model

Definition 1 (Branching step). A parent u holds a committed trunk x_u with $|x_u|=L$. A fan-out is a set of residuals $\{\rho_i\}_{i=1}^k$, $|\rho_i|=\ell_i$. The bound winner $\star$ leaves the spine $x_\star = x_u \parallel \rho_\star$, $|x_\star|=L+\ell_\star$.

Let b be KV bytes per token and P the page size. The three occupancies of one step are

$$\begin{aligned} M_{\text{clone}} &= b(kL + \sum_i \ell_i), \\ M_{\text{CoW}} &= b(L + \sum_i \ell_i), \\ M_{\text{spine}} &= b(L + \ell_\star). \end{aligned} \quad (2)$$

M_{clone} is the first-expand footprint of recompute, APC, and baseline disaggregation; M_{CoW} is one shared trunk with every residual live; M_{spine} is the footprint after abort. W is the token count issued to a prefill kernel before $\star$ is known; X is the token count inserted on the P/D connector. Table 2 lists the remaining symbols.

3 DESIGN

ForkServe lies between the agent runtime and the LLM engine (Figure 1). The harness exposes branch structure; the

control plane admits, prunes, binds, and retains; the backend moves pages and tokens.

3.1 API

<code>open(session)</code>	Create a root; prefill system+user.
<code>fork(u, bid, tok?)</code>	Child of u ; optional known suffix.
<code>speculate(v, pri)</code>	Mark v eligible for speculative prefill.
<code>commit(u, tokens)</code>	Bind by LCP; start decode.
<code>join(children, pol)</code>	Merge fan-out (all/first/kofn).
<code>abort(v) / close</code>	Drop a subtree; end the session.

`commit` is the only operator on the critical path of user-visible tokens.

3.2 Copy-on-Write Context Tree

Definition 2 (Context tree). A session is a rooted tree $T = (V, E)$. Node u stores a token sequence x_u such that $x_{\pi(u)} \preceq x_u$, a residual $\delta_u = x_u[|x_{\pi(u)}|:]$, a mode $\rho_u \in \{\text{SPEC}, \text{COMMIT}, \text{IDLE}, \text{DEAD}\}$, a generation g_u , and a logical page table $\sigma_u = (\text{parent}, \text{alias_len}, \rho_u^{\text{pages}})$.

Invariants: (i) prefix; (ii) at most one committed child per node except during join; (iii) speculative nodes are invisible to sampling; (iv) a page’s refcount equals the number of non-dead nodes that map it. `fork(u, bid, xk)` copies two words ($\text{parent} \leftarrow \sigma_u$, $\text{alias_len} \leftarrow |x_u|$) and increfs the aliased frames— $O(1)$ in L . A write that hits $\text{ref} > 1$ or a read-only tail CoW-copies only the valid rows. Figure 2 is that pool. A multi-level fan-out is a *sequence* of CoWs (Figure 4): each edge aliases the parent and allocates only $+r$; live paths share every page on the common prefix; abort is residual-only, and a subtree with no live child cascades until the committed spine. The two compose: LCP commit publishes full winner pages into the APC index so a later session can hash-hit them (§3.4). Abort frees $b\ell_i$ bytes, not $b(L + \ell_i)$. TTL and offload are node-granular: speculative leaves first, then idle committed leaves, then the spine [20, 43].

3.3 Admission and Commit

A candidate of parent u is a pair $(x^k, \hat{x}^o)$ with admit probability p and prefix-hit rate q . Known suffixes ($p=q=1$)

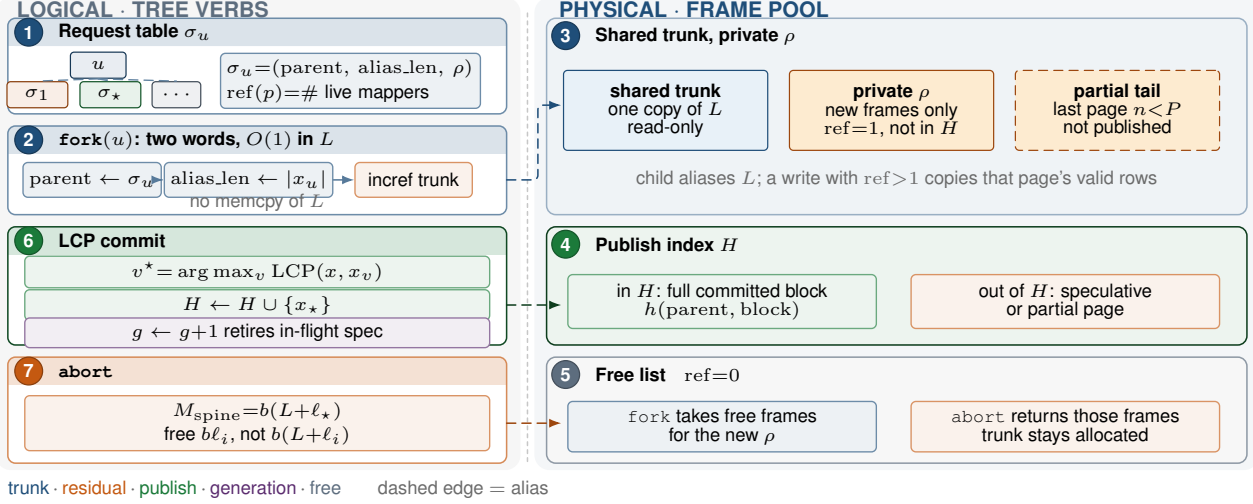


Figure 2. **Context-tree page pool.** `fork` aliases the trunk in $O(1)$ and allocates only ρ ①–③. LCP commit publishes x_* into H ④⑥. `abort` returns residual frames only, so live occupancy is the spine $M_{\text{spine}}=b(L+\ell_*)$ ⑤⑦.

precede observation residuals ($q < 1$), subject to $T_{\text{idle}}(u)$, residual HBM, and branch cap m . Admission stores a future prompt prefix rather than draft output tokens [18, 7].

LCP commit. On prompt x at u ,

$$v^* = \arg \max_v \text{LCP}(x, x_v), \quad g_u \leftarrow g_u + 1. \quad (3)$$

Pages of the LCP become committed; the unmatched tail is a committed prefill; every sibling aborts. A job tagged $g \neq g_u$ drops in $O(1)$. Committed decode attends only KV(x_{v^*}), so the emitted tokens match a baseline that prefills committed tokens only.

3.4 Admission Rule

One action a_i fixes the pair (W_i, X_i) for residual ρ_i (Figure 3). a_i reads m_i first, because a radix match costs no model FLOPs. On a miss it reads $i \in \Lambda$, because a repeated loop is known before a kernel. Only a miss outside Λ enters the kernel, and F_i stops that kernel at $\lceil \alpha \ell_i \rceil$. X_i equals W_i on $\mathcal{K}$ and is zero otherwise, so the connector moves the unmatched suffix of retained residuals. Index 0 is in $\mathcal{K}$; pages stay out of H until (3). The alias of §3.2 is what removes L from both quantities. On Qwen3-14B GSM8K the alias already stores less KV than APC, while median fan-out is 140 ms against 87 ms because every residual is still prefilled; a_i is the term that removes that prefill.

Action. Let H be the published radix index and P the page size. The matched prefix, the loop set, and the early-

abort bit are

$$m_i = \max\{jP : jP \leq \ell_i, \rho_i[0:jP] \in H\}, \quad (4)$$

$$\Lambda = \{i \neq 0 : \text{Loop}(\rho_i)\},$$

$$F_i = \lceil s(\rho_i[0:\lceil \alpha \ell_i \rceil]) \rceil < \tau.$$

The trunk is the alias $\sigma_i \leftarrow (\sigma_u, |x_u|)$, so $L \notin W_i$. With $\mathcal{A} = \{\text{HS}, \text{HP}, \text{DS}, \text{EA}, \text{PF}\}$ and first-match Iverson brackets,

$$a_i = \text{HS}[m_i = \ell_i] + \text{HP}[0 < m_i < \ell_i] + \text{DS}[m_i = 0][i \in \Lambda] + \text{EA}[m_i = 0][i \notin \Lambda][F_i] + \text{PF}[m_i = 0][i \notin \Lambda][\neg F_i]. \quad (5)$$

Index 0 is forced into $\mathcal{K}$, and $a_0 \in \{\text{HS}, \text{HP}, \text{PF}\}$.

$$W_i = \lceil \alpha \ell_i \rceil [a_i = \text{EA}] + (\ell_i - m_i)[a_i = \text{HP}] + \ell_i[a_i = \text{PF}], \quad (6)$$

$$X_i = W_i[i \in \mathcal{K}][a_i \in \{\text{HP}, \text{PF}\}]. \quad (7)$$

Figure 5 is (5)–(7) on one radix tree: green pages lie in H , orange pages are grown, gray pages are withheld.

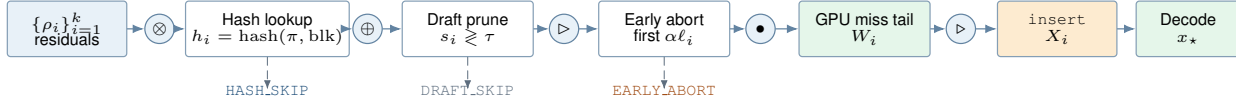
$$d_i^\times = D \cdot [i = 0 \vee s_i \geq \tau]. \quad (8)$$

(8) drops every $s_i < \tau$ before prefill. Algorithm 1 keeps DS only for $i \in \Lambda$; every other low score is prefilled and probed. Table 3 is the resulting first-expand span.

Lemma 1 (First-fan-out dominance). *On a cold index, $W_{\text{APP}} \leq W_{\text{CoW}} \leq W_{\text{APC}} = W_{\text{PD}}$ and $X_{\text{APP}} \leq X_{\text{PD}}$, with strict inequality in W whenever at least one repeated loop is draft-skipped, and strict inequality in X whenever $\mathcal{K} \neq [k]$ or any kept residual is a hash hit.*

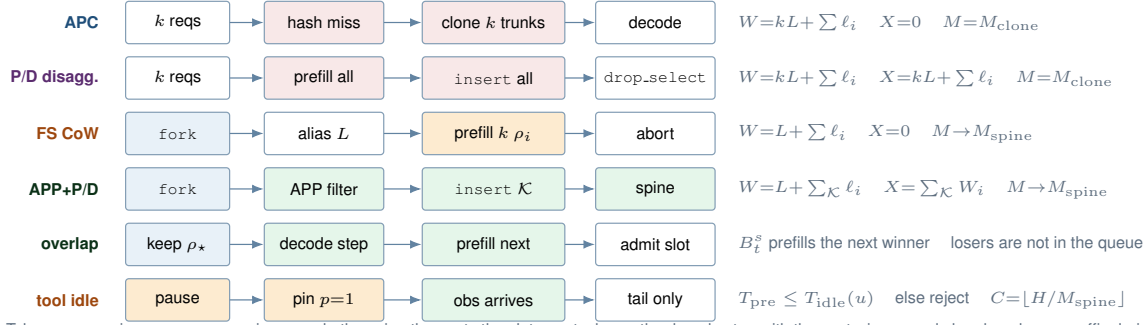
ρ_i : i -th residual h_i : block hash s_i : draft score α : early-abort fraction κ : chunk W_i : GPU miss tail X_i : connector tokens a_i : action

(a) APP + P/D pipeline (Alg. 1)



Parallelism: k residuals Layout: location-addressed CoW, not content-addressed APC Winner $i=0$ is never dropped Spec pages unpublished
Work $W = \sum_i W_i$ Transfer $X = \sum_{i \in \mathcal{K}} W_i$ Occupancy after abort $M_{\text{spine}} = b(L + \ell_*)$ Replay of x_* : $W = X = 0$

(b) First-fan-out dataflow (same k, L, ℓ). Bottleneck in red.



Takeaway: pruning removes non-winner work; the spine then sets the slot count, shares the decode step with the next winner, and absorbs a known suffix during the tool pause.

Figure 3. The admission rule and where it sits on the path. (a) a_i is the first match of (5). Only W_i for $i \in \mathcal{K}$ reaches insert. (b) APC and baseline disaggregation copy the trunk. The alias removes L from W and from X . The overlap row prefills the next miss during decode. The tool-interval row prefills a suffix with $p=1$ while $T_{\text{pre}} \leq T_{\text{idle}}$.

Table 3. Work, transfer, and memory spans (bottleneck in red). W_{pre} is GPU prefill tokens; X is connector tokens; M is live KV after abort; T_{ab} is abort cost in pages; replay is a later open of the published winner. PAR is the number of trunk copies touched on the first expand.

	W_{pre}	X	M after abort	T_{ab}	PAR _{trunk}	Replay W_{pre}
Recompute	$kL + \sum_i \ell_i$	0	$kL + \sum_i \ell_i$	$\Theta(kL/P)$	k	$kL + \ell_*$
APC (first expand)	$kL + \sum_i \ell_i$	0	$kL + \sum_i \ell_i$	LRU of hashed ρ	k	0
hash prefill (first)	$kL + \sum_i \ell_i$	0	$kL + \sum_i \ell_i$	LRU	k	0
P/D disagg.	$kL + \sum_i \ell_i$	$kL + \sum_i \ell_i$	$kL + \sum_i \ell_i$	full KV transfer	k	0
FS CoW	$L + \sum_i \ell_i$	0	$L + \ell_*$	$\Theta(\ell_i/P)$	1	L_{miss}
FS+ + P/D	$L + \sum_{i \in \mathcal{K}} \ell_i$	$\sum_{i \in \mathcal{K}} W_i$	$L + \ell_*$	lazy, residual only	1	0

$P=16$. Known-suffix items ($p=1$) are admitted before residuals ($p < 1$). Spec pages are never hashed; LCP commit publishes full blocks of x_* .

From the action to the wire. The index key is $\text{hash}(\text{parent}, \text{block_tokens}, \text{extra})$ [17], and only full pages enter H , so m_i in (4) steps by P . (3) is what later sets $m_i = \ell_i$. A hit is already visible on the decoder, hence $X_i = 0$ when $a_i = \text{HS}$. (8) withholds every $s_i < \tau$; Algorithm 1 restricts that withhold to $i \in \Lambda$ and continues any other low score past t_0 iff $s'_i \geq \tau$. The t_0 probe is a decode prefix; F_i is a prefill prefix. Defaults are $\tau_{\text{pre}} = 0.15$, and on MATH-500 $\tau = 0.45$, $t_0 \in \{64, 128\}$. Baseline disaggregation moves (1); the rule moves (7).

$$T_{\text{fan}} = T_{\text{pre}}(W) + \mu X + T_{\text{CoW}} + T_{\text{abort}}. \quad (9)$$

T_{CoW} is a pointer swap. §4.2 evaluates (8) at 139 ms (APC), 162 ms (baseline disaggregation), 47 ms (alias only), and 31.4 ms. §4.2 uses Algorithm 1 and cuts computed prefill by 23%.

3.5 Critical-Path Placement

Abort under the rule leaves $M_{\text{spine}} = b(L + \ell_*)$, so the same pool admits

$$C = \lfloor H/M \rfloor \quad (10)$$

sessions, with $M = M_{\text{spine}}$ for ForkServe and $M = M_{\text{APC}}$ while a prefix cache still holds aborted residuals.

$$\Theta \propto C/T_{e2e}. \quad (11)$$

On tick t the chunk budget [2] splits by the committed rate $\underline{s}_r$ [52],

$$B_t^c = \min\left(B_t, \sum_{r \in R_t^c} \max(\underline{s}_r \Delta, 1)\right), \quad (12)$$

$$B_t^s = B_t - B_t^c. \quad (13)$$

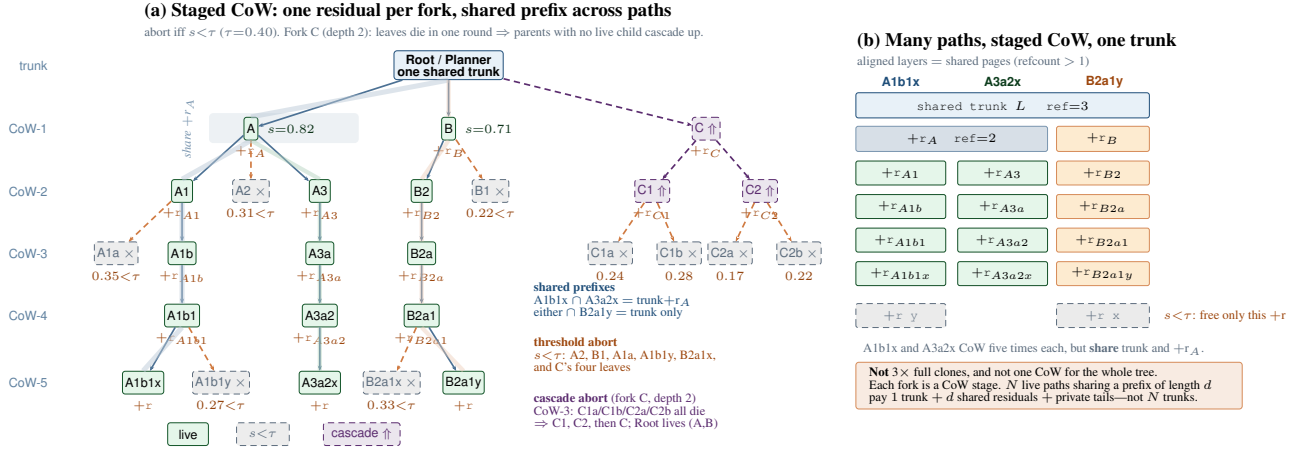


Figure 4. Staged CoW along many paths, threshold abort, cascade abort, shared prefixes. (a) Each edge is one CoW: the child aliases the parent’s pages and allocates only a residual (+r). Live paths A1b1x, A3a2x, and B2a1y share the trunk; the first two also share +r_A. Fork C has depth 2: at CoW-3 all four leaves die, so abort cascades through C1 and C2, then C; it stops at Root, which still has live children A and B. (b) KV stacks of the three live leaves, layers aligned: a shared layer is one physical page set with refcount > 1, not N copies. Aborting A1b1y or B2a1x frees only that leaf residual; cascade-aborting C also frees +r_{C1}, +r_{C2}, and +r_C.

Algorithm 1 Probe admission, replacing (8). Λ is the repeated-loop set. y_i is a probe of t_0 tokens and s'_i is its draft score. Iverson brackets select the decode length. DS applies only to Λ .

Input: $\{\rho_i\}_{i=0}^{k-1}$, threshold τ , probe t_0 , budget D
Output: action a_i , decode length d_i , set $\mathcal{K}$

- 1: $\Lambda \leftarrow \{i \neq 0 : \text{Loop}(\rho_i)\}$
- 2: **for** $i = 0$ to $k - 1$ **do**
- 3: $s_i \leftarrow \text{DraftScore}(\rho_i)$
- 4: **if** $i \in \Lambda$ **then**
- 5: $(a_i, d_i) \leftarrow (\text{DS}, 0)$ $\{(8) \text{ set } d_i^x = 0 \text{ for all } s_i < \tau\}$
- 6: **else if** $i = 0 \vee s_i \geq \tau$ **then**
- 7: $(a_i, d_i) \leftarrow (\text{PF}, D)$
- 8: **else**
- 9: $y_i \leftarrow \text{Dec}(x_u || \rho_i, t_0)$
- 10: $s'_i \leftarrow \text{DraftScore}(y_i)$
- 11: $(a_i, d_i) \leftarrow (\text{PF}, D[s'_i \geq \tau] + t_0[s'_i < \tau])$
- 12: **end if**
- 13: **end for**
- 14: $\mathcal{K} \leftarrow \{i : a_i \neq \text{DS}\}$
- 15: $W_i \leftarrow \ell_i \cdot [a_i = \text{PF}]$
- 16: $\text{DecTokens} \leftarrow \sum_{i=0}^{k-1} d_i$

B_t^c decodes the spine. B_t^s prefills the next miss tail, or a known suffix ($p=1$) when $T_{\text{pre}}(|w|) \leq T_{\text{idle}}(u)$. The observation then pays only the unmatched tail under (3). Speculative tokens stay outside the sampler.

4 EVALUATION

The GPU measurements share one node and one engine, and they differ by model, tensor-parallel width, and workload. §4.1 is the full GSM8K forest on Qwen3-14B at

tp=2. §4.2 is the earlier blanket-skip cost model of APP, with no GPU, against APC, hash prefill, baseline prefill–decode disaggregation, ForkServe, and three decoding-time pruners; §4.2 supersedes that predicate. §4.2 measures the GSM8K prefill threshold $\tau_{\text{pre}}=0.15$ on Qwen3-8B and the accuracy–latency trade-off when that threshold precedes ESC, Speculative Rejection, and DPTS. §4.3 runs ForkServe+ on Qwen3-8B at tp=2 and tp=4. §4.4 is a three-turn Tree-of-Thoughts on Qwen3-4B and Qwen3-8B, one problem per generate. §4.5 repeats the forest on six families at $n=40$. §4.6 sweeps branching, depth, pruning, and batch width on Llama-3-8B, Mistral-7B, and DeepSeek-R1-Distill-Llama-8B, then runs a 2048-token contest forest on that DeepSeek model and on Qwen3-4B.

Setup. GPU runs use vLLM on four A100-SXM4 40GB GPUs, with bfloat16 weights and tp=2 or 4. The models are Qwen3-14B, Qwen3-8B, Qwen3-4B, Qwen2.5-Math-1.5B, Llama-3-8B, Mistral-7B, and DeepSeek-R1-Distill-Llama-8B. Tree-of-Thoughts uses $k=4$ and a 512-token budget (2048 on the contest forest), in batches of eight. The three-turn run in §4.4 issues one problem per generate. A math answer is correct when its final number matches the reference.

4.1 GPU forest

Setup. This section uses Qwen3-14B at tp=2. The GSM8K test set has 1319 problems. Each problem is expanded with four thoughts and decoded for at most 512 tokens, in 165 batches of eight. Accuracy is the fraction of problems whose final number matches the reference.

in H
 prefill
 withheld
 one box = P tokens, shared radix root

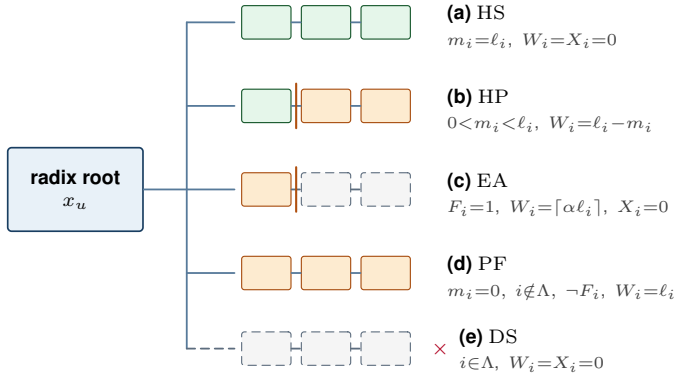


Figure 5. **One radix tree, five actions** ((5)–(7)). The root is the aliased trunk x_u . Each edge is one page of P tokens. Green pages are in H ; orange pages are prefilled; gray pages are withheld. (a) HS: $m_i = \ell_i, W_i = X_i = 0$. (b) HP: $0 < m_i < \ell_i, W_i = \ell_i - m_i$. (c) EA: $m_i = 0, F_i = 1, W_i = \lceil \alpha \ell_i \rceil, X_i = 0$. (d) PF: $m_i = 0, i \notin \Lambda, \neg F_i, W_i = \ell_i$. (e) DS: $i \in \Lambda, W_i = X_i = 0$. A score below τ with $i \notin \Lambda$ is prefilled and probed (Algorithm 1).

Table 4. GSM8K test, Qwen3-14B, $\text{tp}=2, n=1319$. Median over 165 batches of eight. Best storage in green.

	recompute	APC	FS
Accuracy	0.847	0.844	0.845
Peak KV, median (tok)	4136	1430	1030
Peak KV, max (tok)	4944	1632	1232
Peak / APC	$2.89 \times$	$1.00 \times$	$0.72 \times$
Fan-out, median (ms)	326	87	140
Generation (min)	—	18.6	18.9
KV saving vs. rec.	—	0.65	0.75

Storage and expand latency. Table 4 reports the storage result and the expand-latency gap that APP is designed to close. Median peak KV is $0.72 \times$ APC and $0.25 \times$ recompute, and accuracy stays within one point of both. Storage and fan-out do not share that ordering: median fan-out is 140 ms for ForkServe and 87 ms for APC, because copy-on-write and abort execute on the expand path, whereas an APC hash hit does not. End-to-end generation remains within 2% of APC (18.9 vs. 18.6 minutes). The same peak ordering, $0.58\text{--}0.80 \times$ APC, holds on seven math sets for Qwen3-14B, Qwen3-8B, and DeepSeek-R1-Distill-Llama-8B at $\text{tp}=2$ and $\text{tp}=4$. The peak is unchanged when the tensor width doubles; it follows the committed path, as (2) requires.

4.2 Prefill pruning

Setup. This subsection reports the earlier blanket-skip cost model. The GPU admission rule is §4.2 and §4.2. Control plane, no GPU. Ten sessions, $k=4, L=256, \ell=32$. Two of four residuals fail the draft score and are withheld before prefill; the winner is index 0. Prefill time is $\ell \times 12 \mu\text{s}/\text{token}$; P/D transfer is $2 \mu\text{s}/\text{token}$. APC and hash prefill publish

Table 5. Control-plane fan-out (10 sessions, $k=4, L=256, \ell=32$). Bottleneck in red; ForkServe+ in green. Not GPU wall-clock.

Phase	Method	W	T_{fan}	X	peak	skip	prune
fan-out	APC	11520	139.0	0	11520	0	0
	hash	11520	139.0	0	11520	0	0
	P/D disagg.	11520	162.1	11520	11520	0	0
	FS	3840	47.1	0	3840	0	0
	ForkServe+	2592	31.4	32	2880	9	30
	ForkServe+ / APC	−77.5%	−77.4%	—	−75%	—	—

blocks only after a request materializes them, so the first fan-out clones k trunks. Baseline disaggregation uses that prefill and then transfers the live token span. ForkServe aliases the trunk and prefills the k residuals. ForkServe+ applies the residual action under the stricter predicate (8): every $s_i < \tau$ is withheld before the kernel, and the connector carries only $\mathcal{K}$.

Fan-out. Table 5 is the comparison motivated by the GPU latency gap. On the first expand, hash prefill matches APC: the blocks are unpublished, so the hash probe does not remove the cloned trunks. Baseline disaggregation exceeds APC (162 vs. 139 ms) because it transfers that clone. ForkServe removes the clone by aliasing the trunk (3840 prefill tokens, 47 ms). APP then withholds prefill from 30 of the 40 residuals, which reduces T_{fan} to 31.4 ms and the connector payload from 11520 tokens to 32.

Decoding-time pruning. Table 6 runs ESC [21], Speculative Rejection [31], and DPTS [19] on this same fan-out. Each method prefills the shared trunk once. The length of the decode that produces the prune score is the length published for that method. ESC stops after a window of 5

Table 6. Decoding-time pruning on the Table 5 fan-out (10×4 , $L=256$). Decode-to-cut is the prefix that must be generated before a child can be dropped. “On losers” counts that prefix on the two low-score children. Same $12 \mu\text{s}/\text{token}$ clock. Not GPU wall-clock.

	prefill	decode	on losers	peak	T_{cut}
ESC	2560	20480	10240	23040	276.5
Spec. Rejection	2560	10240	5120	12800	153.6
DPTS	2560	4000	2000	6560	78.7
ForkServe+	2592	0	0	2880	31.4

Table 7. MATH-500 level ≥ 4 , Qwen3-8B, $n=32$, $D=1024$. Any-finished is exact match on a branch that ran the full budget. “Drop $s < \tau$ ” withholds every low-score residual before prefill. “Grow t_0 ” is Algorithm 1: loops stay at $d_i=0$; every other low-score residual is probed for t_0 tokens and continues to D only if $s'_i \geq \tau=0.45$. Time is trunk warm plus fan-out, in seconds.

	$k=4$	$k=8$	$k=16$
Full fan-out	12/32, 25.3	15/32, 52.5	20/32, 104.1
Drop $s < \tau$	6/32, 19.1	11/32, 25.6	13/32, 52.1
Skip loops only	11/32, 23.0	17/32, 41.5	16/32, 71.4
Grow $t_0=64$	8/32, 24.0	16/32, 42.0	18/32, 72.5
Grow $t_0=128$	9/32, 24.7	13/32, 43.3	20/32, 72.2

agreeing samples; this fan-out has four children and two low-score answers, so the window stays open and every child is decoded to the 512-token budget. Speculative Rejection is charged its partial-reward horizon $\tau=256$ and then drops the lower half ($\alpha=0.5$); the memory-limit trigger in that paper would not fire at $k=4$, so τ is the earlier checkpoint. DPTS early-stops after its 100-token mini-step. The two low-score children are the ones removed, and the winner is kept. Decoded tokens use the same $12 \mu\text{s}/\text{token}$ clock. DPTS is the earliest of the three and still decodes 4000 tokens before the drop, 2000 of them on the rejected children, reaching the cut at 78.7 ms. Under this cost model APP has already withheld those residuals: 0 tokens on rejected children and $T_{\text{fan}}=31.4$ ms. That row is the blanket skip of (8). §4.2 supersedes it: only a repeated loop is withheld before prefill.

Probe admission. (8) is inadmissible once the residual is not a loop. We measure Algorithm 1 on Qwen3-8B, one GPU each, four A100-40GB, $\text{tp}=1$. The slice is the first 32 MATH-500 problems at level 4 or 5. Temperature is 0, EOS is ignored, prefix caching is on, and the decode budget is $D=1024$. Half of the k residuals are contest strategies. The other half alternates a repeated loop with illegal text. A problem counts when some branch that ran all 1024 tokens matches the reference answer. The winner is residual 0.

Table 7 is the correction. Dropping every residual below τ cuts any-finished from 12/32 to 6/32 at $k=4$ and from 20/32 to 13/32 at $k=16$. Loop branches are 0 for 0 at every k , so skipping them before prefill is safe. At $k=16$, a 128-token probe extends every illegal residual (128/128) and

Table 8. GSM8K, Qwen3-8B, $n=16$, $D=512$, ESC only. Any-finished is exact match on a full-budget branch. $\tau_{\text{pre}}=0.15$ drops the loop; 0.45 also drops illegal text. Time is trunk warm plus fan-out, in seconds. Computed prefill falls 23% at 0.15 and 45% at 0.45 at every k .

	$k=4$	$k=8$	$k=16$
ESC	16/16, 8.65	16/16, 10.84	16/16, 17.05
APP $\tau_{\text{pre}}=0.15$	15/16, 8.22	16/16, 10.19	16/16, 14.14
APP $\tau_{\text{pre}}=0.45$	14/16, 7.89	14/16, 8.60	16/16, 10.93

Table 9. Same GSM8K slice, $k=16$, $\tau_{\text{pre}}=0.15$ in front of three host decoders. Admitted children and computed prefill tokens are identical across hosts: prefill admission does not read τ , d_{DPTS} , or α . Any-finished stays 16/16.

	adm.	prefill	e2e (s)	any
ESC	16	5008	17.05	16/16
ESC+APP	12	3848	14.14	16/16
Spec. Rejection	16	5008	13.14	16/16
SR+APP	12	3848	12.47	16/16
DPTS	16	5008	11.86	16/16
DPTS+APP	12	3848	11.49	16/16

none of the loops. Any-finished returns to the full fan-out, 20/32, while end-to-end time falls from 104.1 s to 72.2 s and decode tokens from 524288 to 393216. Probing the loops as well extends none of them and costs 74.4 s for the same 20/32. At $k=8$ the shorter probe is the one that holds accuracy: grow at $P=64$ is 16/32 in 42.0 s, against 15/32 in 52.5 s for the full fan-out. At $k=4$ the probe does not recover the full fan-out (9/32 against 12/32); skipping the single loop, and decoding the illegal residual to D , is 11/32 in 23.0 s. Figure 6 is Table 7: at $k=16$ grow $t_0=128$ meets the full-fan-out count at 72.2 s against 104.1 s.

Accuracy-speed trade-off. The MATH-500 probe still uses a decode threshold $\tau=0.45$ on the generated prefix. Prefill admission is a different score, with default $\tau_{\text{pre}}=0.15$: it drops only the repeated loop (0.02) and starts illegal text (0.22) and every live strategy (1.0). We measure that rule on Qwen3-8B GSM8K, $n=16$, $D=512$, the same hopeless mix, four A100-40GB, $\text{tp}=1$. The host decoder is ESC, so a withheld child never pays residual prefill or a decode slot. A denser sweep of τ_{pre} collapses to three admission sets; 0.15 and 0.45 are the two nonzero operating points.

Table 8 is the accuracy-latency trade-off. $\tau_{\text{pre}}=0.15$ attains 15/16, 16/16, 16/16 and reduces wall-clock time as the tree widens (−5%, −6%, −17%). Computed prefill tokens fall from 1320/2624/5008 to 1018/2020/3848. Raising τ_{pre} to 0.45 reduces end-to-end time by −9%, −21%, −36%, with 732/1448/2768 computed prefill tokens. Accuracy falls by two problems at $k=4$ and $k=8$. At $k=16$ both thresholds remain at 16/16, so 0.45 is the faster matched-accuracy setting (10.93 s vs. 14.14 s). The default is 0.15 because accuracy holds at every k .

Table 9 stacks the same $\tau_{\text{pre}}=0.15$ in front of each host.

MATH-500, Qwen3-8B, $n=32$, $D=1024$, $\tau=0.45$

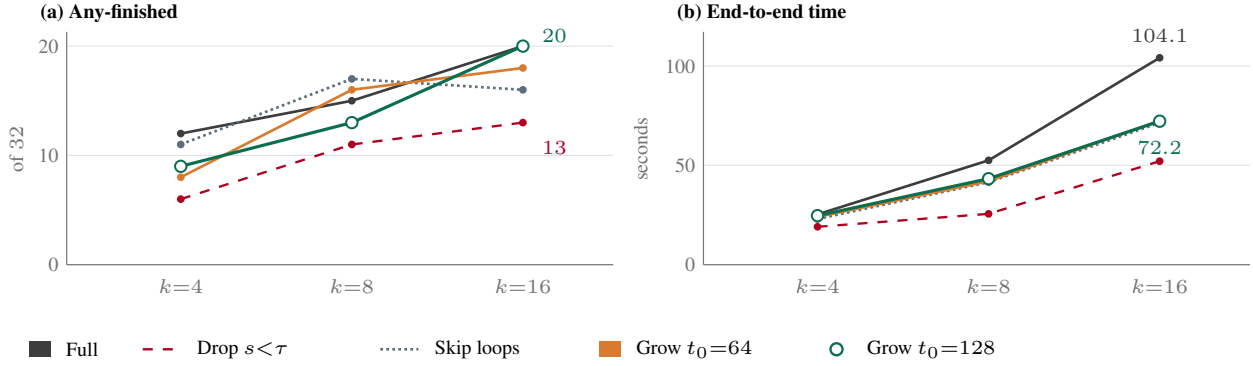


Figure 6. **Probe admission on MATH-500** (Table 7). Any-finished counts a problem only when a full-budget branch matches. Dropping every $s < \tau$ is the fast curve and the low count. Grow $t_0=128$ returns to 20/32 at $k=16$ and cuts 104.1 s to 72.2 s. At $k=8$, grow $t_0=64$ is 16/32 in 42.0 s, above the full fan-out.

ESC, Speculative Rejection, and DPTS alone all report 5008 computed prefill tokens: a decode-time cut still prefills every child. Adding APP admits 12 of 16 residuals and 3848 prefill tokens on every host. Wall-clock falls 17% on ESC, 5% on Speculative Rejection, and 3% on DPTS, because those last two hosts would have stopped the loop later anyway. The accuracy that Table 8 preserves is unchanged: every row in Table 9 is 16/16. At $k=8$ the same threshold is 16/16 on ESC (10.84 to 10.19 s) and lifts Speculative Rejection and DPTS from 15/16 to 16/16. Figure 7 plots these rows. Prefill admission is identical across the three hosts: 16 children and 5008 tokens become 12 and 3848.

Concurrency from measured KV. Our empirical results on DeepSeek-R1 GSM8K (761 APC tokens, 561 ForkServe tokens, $n=16$, $k=4$) in an 8 GiB pool in (11) give 76 concurrent trees for APC and 103 for ForkServe. Figure 8 sweeps offered QPS: under a 1 s P99 TTFT constraint the same model admits 384 QPS for APC and 512 for the pruned tree, and saturated decode throughput rises from 48640 to 65920 tokens/s (+35.5%). This uses measured M , not the control-plane peak of Table 5.

4.3 GPU ForkServe+

Setup. Qwen3-8B, bfloat16 , $n=80$, $D=512$, $\text{tp}=2$ and 4 on four A100-SXM4 40GB GPUs. ForkServe+ is the admission rule of §3.4 on the aliased trunk. GSM8K and Game24 are ToT ($k=4$); HumanEval is ReAct wrap+recovery ($k=2$).

Peak KV on this forest is 1182 against APC 1582 ($0.75\times$) at both widths. Game24 and HumanEval peaks are $0.62\times$ and $0.96\times$ APC.

Table 10. Qwen3-8B GPU forest, $n=80$, $D=512$. Peak KV is independent of tp . Game24 / HumanEval peaks stay with the spine.

	peak	D^+	e2e	acc.	time reduction	G24	HE
APC $\text{tp}=2$	1582	40960	36.0	0.900	—	1289	1970
ForkServe+ $\text{tp}=2$	1182	13356	28.2	0.863	$0.78\times$	793	1882
APC $\text{tp}=4$	1582	40960	22.7	0.938	—	1289	1970
ForkServe+ $\text{tp}=4$	1182	12826	16.9	0.925	$0.75\times$	793	1882

Table 11. Three-turn ToT, $n=16$, $k=4$, $\text{tp}=2$, one problem per generate. End-to-end time is seconds per problem. Peak KV is tokens on the live spine. FS is ForkServe; FS+ is ForkServe+. Shortest end-to-end time and lowest peak in green.

	Qwen3-4B			Qwen3-8B		
	APC	FS	FS+	APC	FS	FS+
GSM8K acc.	0.938	0.938	0.938	1.000	0.938	0.938
GSM8K e2e (s)	1.66	1.76	1.63	2.48	2.56	1.91
GSM8K peak	318	275	275	318	275	275
Game24 acc.	0.375	0.500	0.562	0.562	0.562	0.625
Game24 e2e (s)	1.61	1.64	1.13	2.15	2.15	1.52
Game24 peak	265	216	216	265	216	216

4.4 Multi-turn Tree-of-Thoughts

Setup. Qwen3-4B and Qwen3-8B, bfloat16 , $\text{tp}=2$, temperature 0, $n=16$. Each problem is one session of $k=4$ agents and three turns, and each system issues that problem as its own generate. Turns 1 and 2 decode a 32-token step; the losers abort; the winner spine is the parent of the next fan-out. Turn 3 is the published answer, $D=512$ on GSM8K and $D=256$ on Game24.

Occupancy across turns. Table 11 and Figure 10 are GSM8K and Game24 under that schedule. Peak KV is 275 against APC 318 on GSM8K ($0.86\times$) and 216 against 265 on Game24 ($0.82\times$), on both models. ForkServe stays within 6% of APC end-to-end ($1.00\text{--}1.06\times$). Across 16

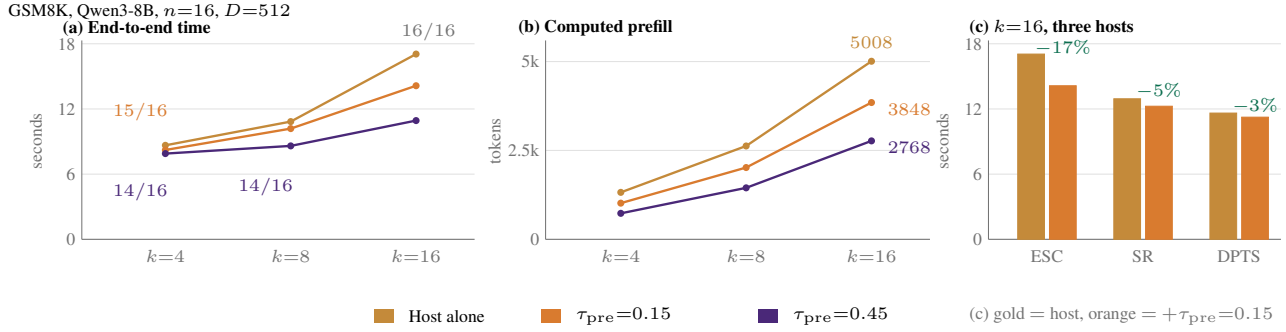


Figure 7. Prefill threshold on GSM8K (Tables 8 and 9). (a) Wall-clock under ESC. Labels are any-finished. $\tau_{\text{pre}}=0.45$ is 14/16 at $k=4$ and $k=8$, and 16/16 at $k=16$. (b) Computed prefill tokens, -23% at 0.15 and -45% at 0.45. (c) The same 0.15 threshold in front of each host at $k=16$. Every row stays 16/16.

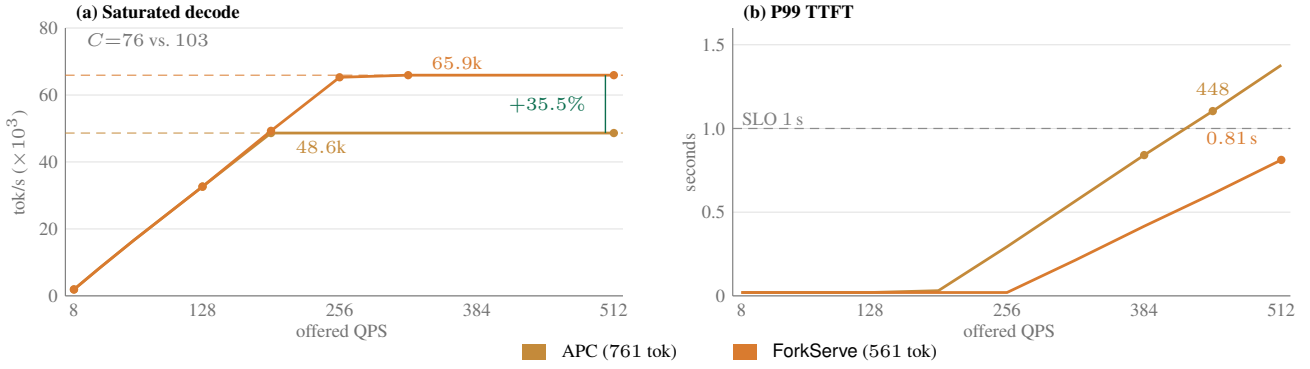


Figure 8. SLO sweep from the measured DeepSeek-R1 GSM8K peaks (761 APC tokens, 561 spine tokens, $n=16$, $k=4$). An 8 GiB pool at 147456 bytes/token holds 76 and 103 trees. Each slot serves 640 tokens/s, so offered load saturates at 48640 and 65920 tokens/s. Under the same model the 1 s P99 TTFT line is crossed by APC at 448 QPS and is still met by ForkServe at 512 QPS (0.81 s).

Table 12. Expand versus decode, milliseconds per problem. Expand is the prefill of the k residuals across three turns. On APC that timer also includes the two 32-token step decodes; ForkServe and ForkServe+ charge those steps to decode. Speedup in Figure 11 is APC expand time over the other system. Shortest time in each trio in green.

	Expand (ms)			Decode (ms)		
	APC	FS	FS+	APC	FS	FS+
4B GSM8K	429	83	55	1215	1680	1572
4B Game24	435	87	53	1166	1552	1074
8B GSM8K	597	94	58	1867	2468	1848
8B Game24	598	94	56	1538	2060	1466

problems, three turns, and four agents, the expand issues 192 residuals. ForkServe+ prefills 48 of them and withholds the other 144 before they reach the GPU.

Table 12 and Figure 11 separate that expand from the winner decode. The expand speedup is 5.0–6.4 $\times$ for ForkServe and 7.9–10.6 $\times$ for ForkServe+. Decode does not match it.

ForkServe decode is slower than APC because the two step decodes are timed there, which is why end-to-end stays at 1.00–1.06 $\times$. GSM8K accuracy matches APC on Qwen3-4B (15/16 on the same problems) and is one problem lower on Qwen3-8B (15/16 vs. 16/16). On Game24, ForkServe solves two more problems than APC on Qwen3-4B (8/16 vs. 6/16) and the same nine on Qwen3-8B; ForkServe+ solves 9/16 and 10/16.

4.5 Cross-model forest

Setup. Same four systems, $n=40$, $D=512$, $\text{tp}=2$ and 4, on Qwen3-4B, Qwen2.5-Math-1.5B, Qwen3-14B, Llama-3-8B-Instruct, Mistral-7B-Instruct, and DeepSeek-R1-Distill-Llama-8B.

Peak occupancy. APC retains the shared trunk and the union of the residuals, which is the copy-on-write term $M_{\text{APC}} = M_{\text{CoW}} = b(L + \sum_i \ell_i)$ from (2). The tree retains only the committed spine $M_{\text{spine}} = b(L + \ell_*)$. The ratio of

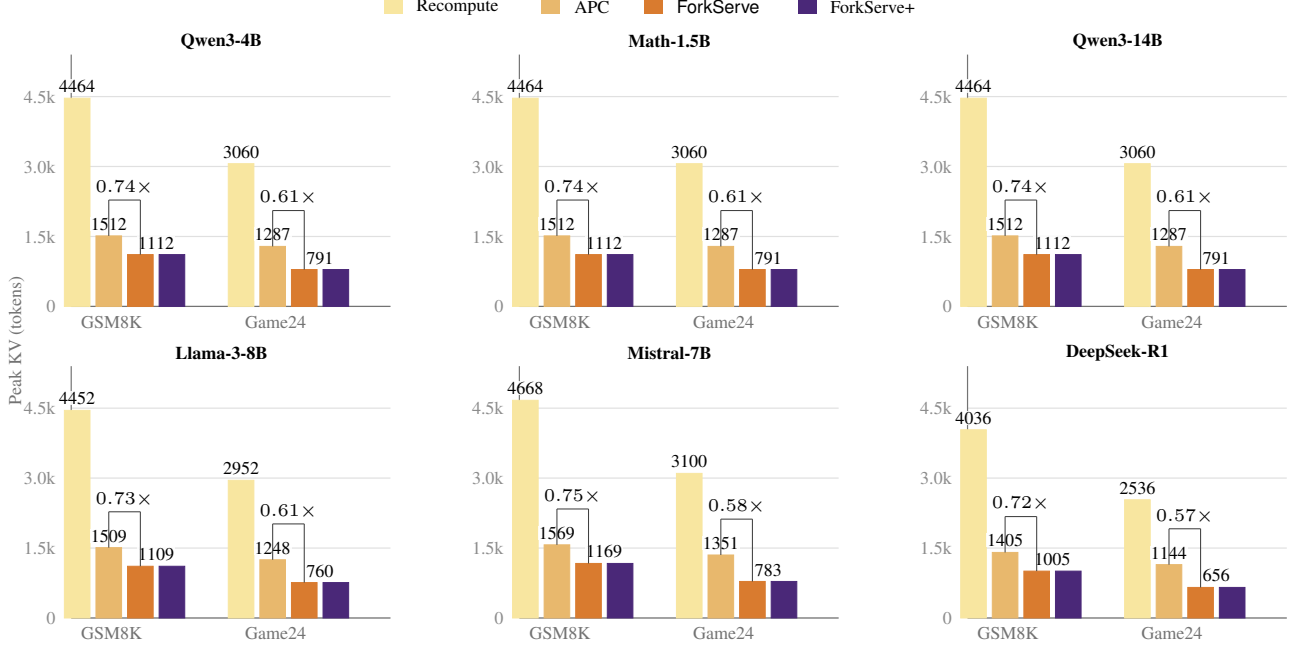


Figure 9. **Peak KV** on six models ($n=40$, $tp=2$). Each panel is one family; groups are GSM8K and Game24. Bars are recompute, APC, ForkServe, and ForkServe+ (cream / tan / orange / purple). ForkServe and ForkServe+ share the spine. Annotated $\times$ is $M_{\text{spine}}/M_{\text{APC}}$: 0.72–0.75 $\times$ on GSM8K and 0.57–0.61 $\times$ on Game24, independent of checkpoint size, as (2) predicts.

these two terms cancels b and equals $(L + \ell_*)/(L + \sum_i \ell_i)$, so it depends on k and the lengths (L, ℓ_*) rather than on the weights or the tensor width. When the residuals have similar length, the ratio simplifies to $(L + \ell)/(L + k\ell)$ and decreases as L/ℓ decreases. Figure 9 shows this pattern: a long trunk on GSM8K gives 0.72–0.75, and a shorter trunk on Game24 gives 0.57–0.61. Figure 12 repeats the comparison at $n=80$ with recompute and ForkServe+ on separate axes. Relative to recompute, which stores the cloned occupancy M_{clone} , the spine is 0.25 $\times$. ForkServe+ matches that ratio. Accuracy remains at the APC level, which indicates that the lower peak is a storage effect on the same committed answer. Figure 13 shows the per-item decode-length distribution on GSM8K and Game24.

HumanEval uses $k=2$, so the residual is a smaller fraction of the context and M_{spine} remains close to M_{APC} . Pass@1 is unchanged across recompute, APC, and ForkServe.

4.6 Branching, pruning, and long traces

Setup. GSM8K, $n=16$, $D=512$, $tp=2$, on Llama-3-8B, Mistral-7B, and DeepSeek-R1-8B. The sweep changes k or adds a second ToT level. MATH-500 ($n=200$) and AIME ($n=73$) use $D=2048$.

The ratio falls with k . Table 14 and Figure 14 measures $(L + \ell)/(L + k\ell)$ on the GPU. The spine peak is invariant in k

Table 13. Last-line answer stop, $n=32$, $tp=2$. ForkServe+ stops only on a finished answer in the last non-empty line. Peak is the max over chunks. GSM8K $D=512$; Game24 $D=256$. R1-8B is DeepSeek-R1-Distill-Llama-8B; L8B is Llama-3-8B; M7B is Mistral-7B. Best storage in green.

Model	Metric	APC	FS+
R1-8B	GSM8K peak	1493	1093
	GSM8K acc.	0.500	0.562
	GSM8K D^+	16384	12693
	Game24 peak	1064	616
L8B	GSM8K peak	1485	1085
	GSM8K acc.	0.719	0.688
	GSM8K D^+	16384	5101
	Game24 peak	1224	736
M7B	GSM8K peak	1521	1121
	GSM8K acc.	0.344	0.188
	GSM8K D^+	16384	6723
	Game24 peak	1296	728

while the APC peak increases, so the ratio moves from 0.89–0.90 at $k=2$ to 0.53–0.55 at $k=8$ on all three families. At depth 2 the ratio is 0.75–0.76. ForkServe+ leaves the peak unchanged. At $k=8$ it rejects 28 branches and reduces fan-out from 307 to 179 ms on Llama-3-8B, from 349 to 191 ms on Mistral-7B, and from 363 to 194 ms on DeepSeek-R1. With draft pruning disabled, Llama-3-8B fan-out is 247 ms and DeepSeek-R1 fan-out is 271 ms. A batch of 16 keeps the ratio at 0.72–0.73 (2071/2871 on Llama-3-8B, 2135/2935 on Mistral-7B, 2093/2893 on DeepSeek-R1).

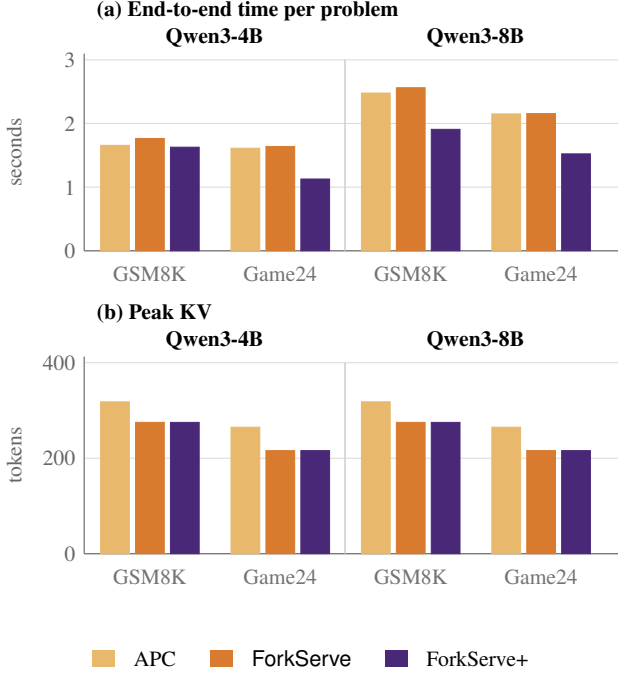


Figure 10. **Three-turn ToT** on Qwen3-4B and Qwen3-8B ($n=16$, $tp=2$, one problem per generate). (a) End-to-end seconds per problem; a shorter bar is faster. (b) Peak KV tokens. ForkServe stays with APC on latency and stores the spine ($0.86\times$ APC on GSM8K, $0.82\times$ on Game24). ForkServe+ is faster because it prefills one of four residuals.

At $D=2048$ the ratio is $0.75\text{--}0.91$ (Table 15). On MATH-500, ForkServe matches APC (432 s vs. 429 s on DeepSeek-R1). ForkServe+ cuts DeepSeek-R1 from 429 s to 247 s and Qwen3-4B from 329 s to 236 s; the Qwen3-4B $n=32$ slice is 38 s versus 53 s. ForkServe+ cuts MATH-500 fan-out from 2406 to 1284 ms on DeepSeek-R1 and from 2787 to 1014 ms on Qwen3-4B, against APC at 1593 and 1201 ms. Qwen3-4B ForkServe+ scores 120/200 on MATH-500 (APC 124/200).

5 RELATED WORK

Request-level serving. Orca [47] introduced iteration-level scheduling; vLLM [17] paged the KV cache and hashes full blocks for APC; SGLang [53] added radix prefix sharing; Sarathi [2, 3] chunked prefills; DistServe and Splitwise [54, 32] disaggregated prefill from decode. Prefix caching discovers sharing after tokens exist, and prefill-decode disaggregation transfers every page the prefiller produced. (5) uses the hash as m_i and the connector as X_i : both are instances of one residual action.

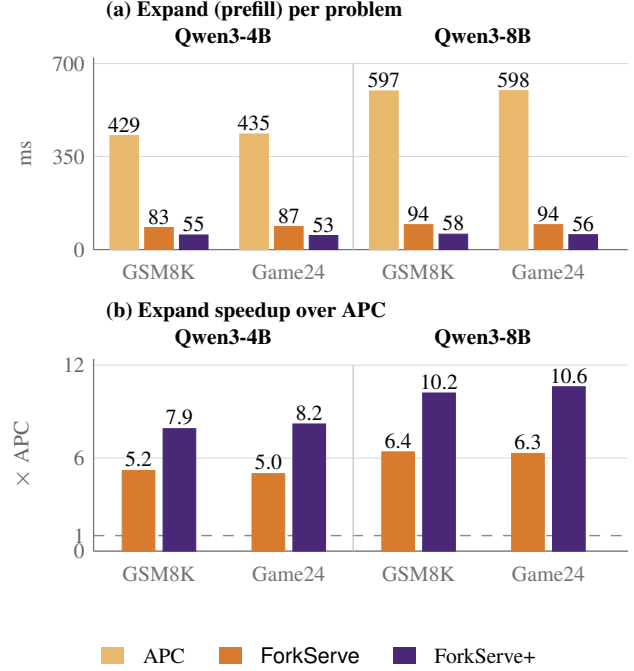


Figure 11. **Expand versus APC** on the same four runs as Table 12. (a) Expand milliseconds per problem. (b) Speedup of that expand over APC. The dashed line is $1\times$. ForkServe is $5.0\text{--}6.4\times$ and ForkServe+ is $7.9\text{--}10.6\times$. Decode in Table 12 does not repeat that ratio: the step decodes sit in the ForkServe decode column, so ForkServe end-to-end stays with APC.

Program- and SLO-aware scheduling. Autellix [25] schedules by program-level attained service; JITServe [52] allocates just-enough bandwidth under heterogeneous SLOs. Hermes [24], ThunderAgent [16], and SAGA [13] raise the scheduling unit to the program or workflow. These systems leave KV unshared across children and leave unissued children unpre-filled.

KV retention and overlap. InferCept [1], Continuum [20], MORI [43], and ConServe [11] keep the *current* context alive or placed. Sutradhara [6] overlaps tool execution with prefill of one linear tool-independent prefix. SmoothAgent [30] lookahead-transforms context; IdleSpec [9] drafts plans for accuracy, not KV. Classical speculative decoding [18, 7, 22] drafts *output* tokens. ForkKV [38] copy-on-write splits LoRA residuals. (5) likewise aliases the shared prefix, then sets W_i and X_i from the unmatched suffix.

Decoding-time pruning. A second line drops a candidate after the model has generated it. Beam search in Tree-of-Thoughts [45] scores a finished step and keeps the top beams. Early-stopping self-consistency (ESC) [21] draws a window

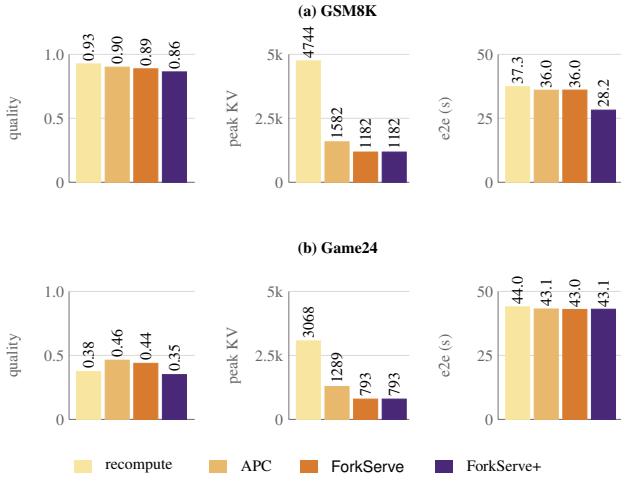


Figure 12. Four systems on Qwen3-8B ($n=80$, $tp=2$, $D=512$). Each panel has its own axis. Quality is higher-better; peak KV and end-to-end time are the measured values, so a shorter bar is better. (a) GSM8K. (b) Game24. Figure 12 shows that ForkServe and ForkServe+ lower peak KV below APC on both workloads ($0.75\times$ on GSM8K, $0.62\times$ on Game24) while GSM8K quality stays with APC, and that ForkServe+ also lowers GSM8K end-to-end time from 36.0 s to 28.2 s.

Table 14. GSM8K peak KV versus branching ($n=16$, $tp=2$, $D=512$). The spine peak is invariant in k ; the APC peak increases with k . ratio is FS/APC. L8B is Llama-3-8B; M7B is Mistral-7B; R1-8B is DeepSeek-R1-Distill-Llama-8B. Best storage in green.

Model/ratio	Method	$k=2$	$k=4$	$k=8$	depth 2
L8B	APC	610	750	998	814
	FS	550	550	550	614
FS/APC		0.90	0.73	0.55	0.75
M7B	APC	629	769	1033	841
	FS	569	569	569	641
FS/APC		0.90	0.74	0.55	0.76
R1-8B	APC	629	761	1053	841
	FS	561	561	561	641
FS/APC		0.89	0.74	0.53	0.76

of complete samples and skips later samples only once that window agrees. Speculative Rejection [31] ranks partial generations and halts the lower quantile. Dynamic Parallel Tree Search (DPTS) [19] generates a fixed mini-step on every active node, then early-stops low-confidence children. In each method the score is computed from tokens already resident, so the prefill and a decode prefix precede the drop. (5) withholds $i \in \Lambda$ before that prefill and continues every other low score past t_0 only if $s'_i \geq \tau$ (§4.2).

6 CONCLUSION

Disaggregated prefill transfers every computed page, and a prefix cache retains aborted residuals until eviction. ForkServe replaces both with one action per residual: prefill

Table 15. Contest forest, $D=2048$, $k=4$, $tp=2$. MATH-500 is $n=200$; AIME is $n=73$. R1 is DeepSeek-R1; 4B is Qwen3-4B. Best storage and shortest end-to-end time in green.

Model	Method	peak	/ APC	acc.	e2e (s)
R1 MATH	APC	1863	—	0.560	429
	FS	1391	0.75	0.510	432
	FS+	1599	0.86	0.545	247
R1 AIME	APC	2118	—	0.055	178
	FS	1646	0.78	0.123	178
	FS+	1662	0.78	0.096	165
4B MATH	APC	1937	—	0.620	329
	FS	1770	0.91	0.625	248
	FS+	1770	0.91	0.600	236
4B AIME	APC	2150	—	0.027	136
	FS	1794	0.83	0.041	136
	FS+	1794	0.83	0.041	131

and transfer equal the suffix absent from the radix index, a repeated loop is omitted, and a score below τ continues past t_0 tokens only when the generated prefix meets τ . Because the trunk is an alias, that suffix excludes L , and abort leaves the spine $b(L + \ell_*)$.

On Qwen3-8B GSM8K ($n=16$, $D=512$), $\tau_{pre}=0.15$ ahead of ESC, Speculative Rejection, and DPTS attains 15/16, 16/16, 16/16 at $k=4, 8, 16$, cuts computed prefill by 23% (5008 to 3848 at $k=16$), and reduces ESC time from 17.05 s to 14.14 s at $k=16$. $\tau_{pre}=0.45$ reaches 10.93 s at $k=16$ with the same 16/16 and loses two problems at $k=4$ and $k=8$. On MATH-500 ($n=32$, $D=1024$), withholding every score below τ drops any-finished accuracy from 20/32 to 13/32 at $k=16$; a 128-token probe restores 20/32 and lowers end-to-end time from 104.1 s to 72.2 s. The Qwen3-14B spine is $0.72\times$ the median APC peak at matched accuracy (0.845 vs. 0.844). Across six families the ratio stays in $0.72\text{--}0.75\times$ and falls from $0.89\text{--}0.90\times$ at $k=2$ to $0.53\text{--}0.55\times$ at $k=8$. At a 2048-token budget it remains $0.75\text{--}0.91\times$.

The same action sets the kernel, the connector payload, and the KV that remains. Occupancy depends on k and residual length, not on tensor-parallel width. As fan-out, depth, and context grow against a fixed pool, each new branch aliases the trunk and abort returns only its residual.

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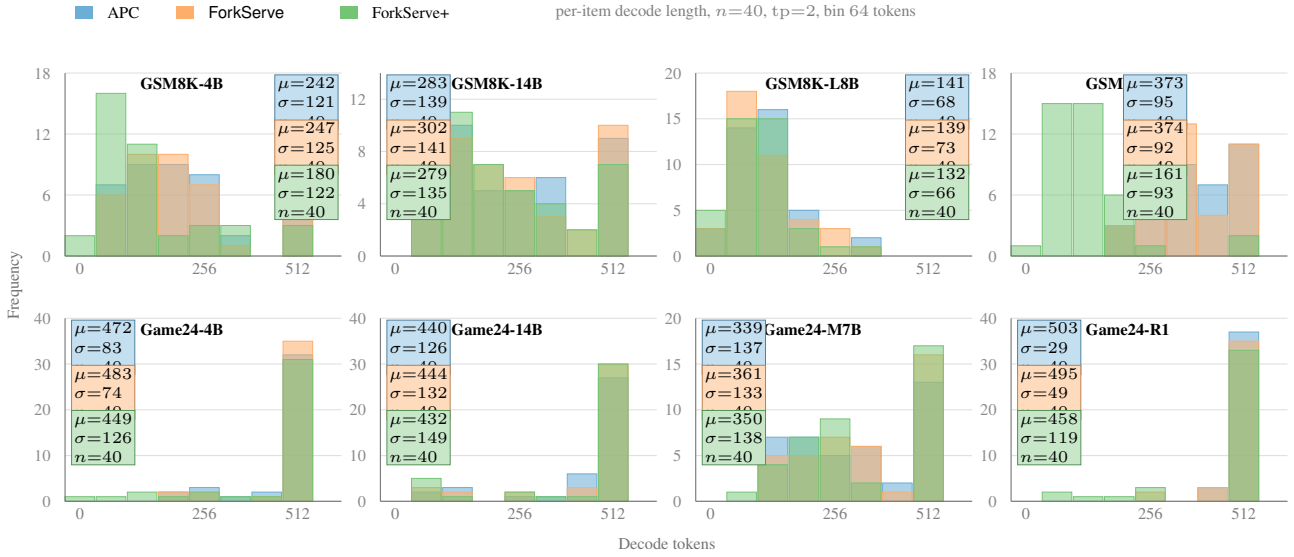


Figure 13. Per-item decode-length histograms ($n=40$, $tp=2$, $D=512$). Each panel overlays APC, ForkServe, and ForkServe+ (bin width 64). Boxes report μ , σ , and n . GSM8K (top) and Game24 (bottom) overlay APC, ForkServe, and ForkServe+. Llama-3-8B Game24 is omitted (all lengths 1); Mistral-7B is shown instead.

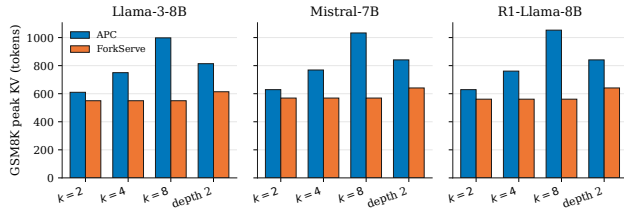


Figure 14. GSM8K peak KV versus branching ($n=16$, $tp=2$, $D=512$). ForkServe+ matches the ForkServe bar. The spine peak is invariant in k ; the APC peak grows with the hashed union, so the ratio falls from 0.89–0.90 to 0.53–0.55.

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